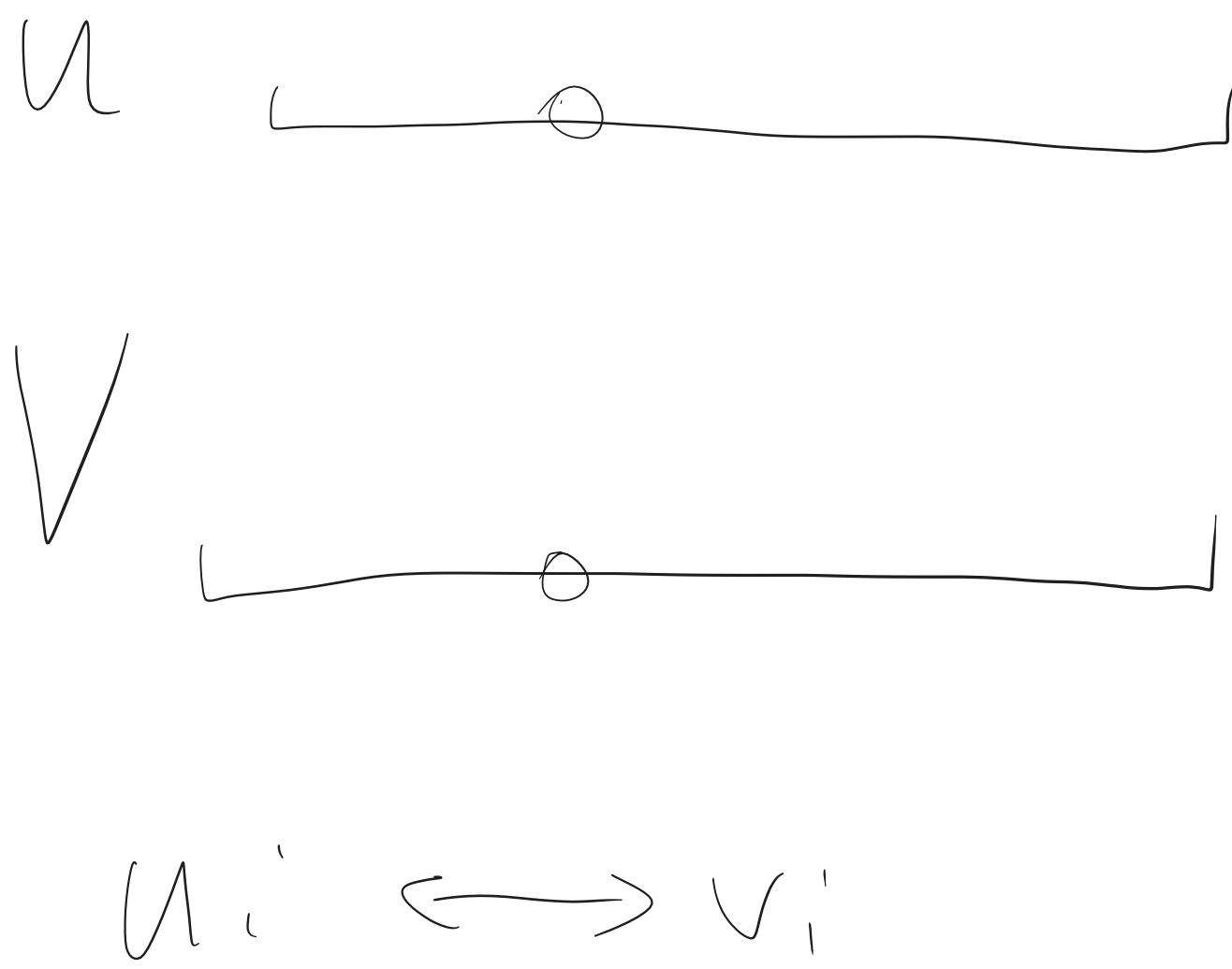
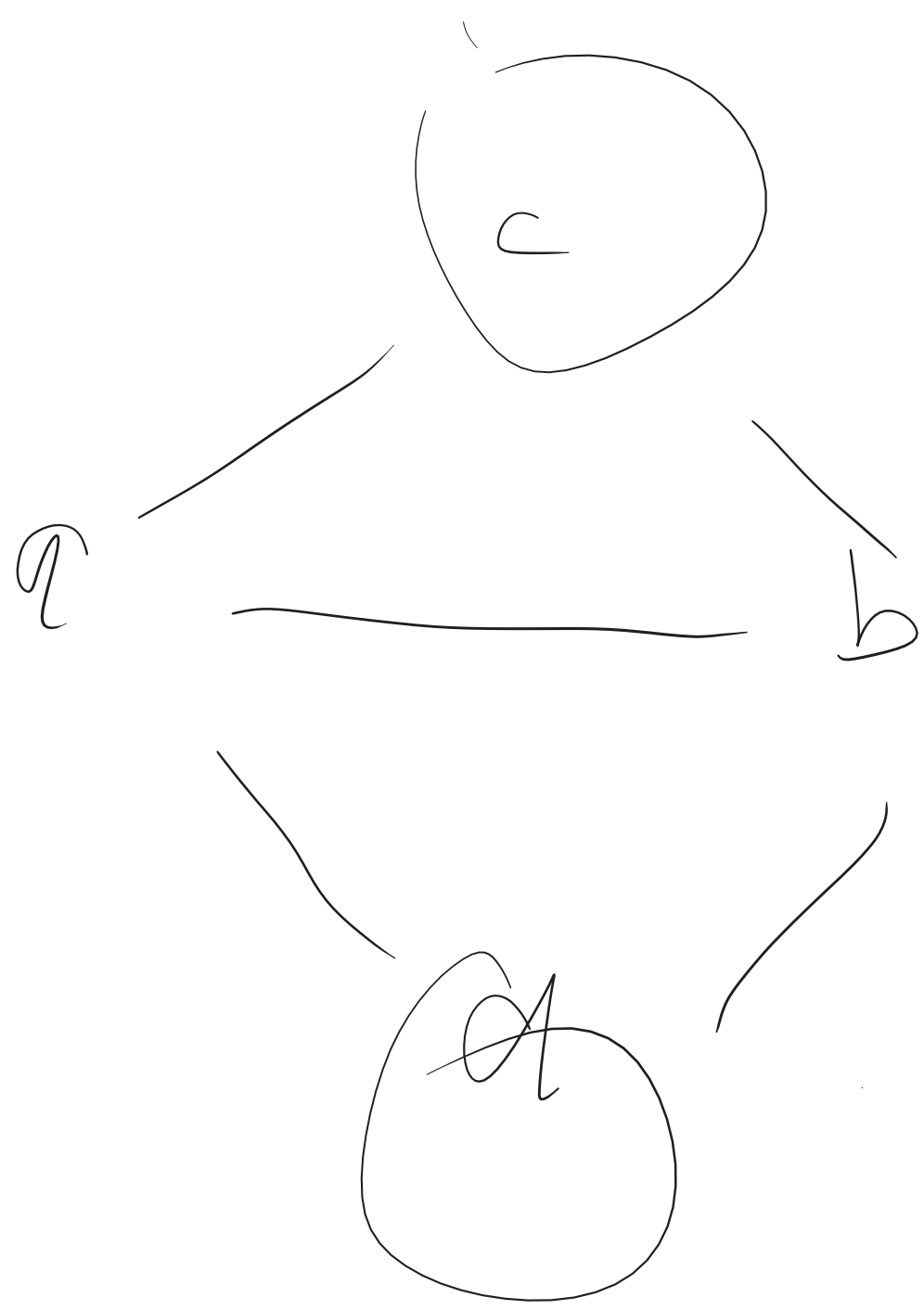
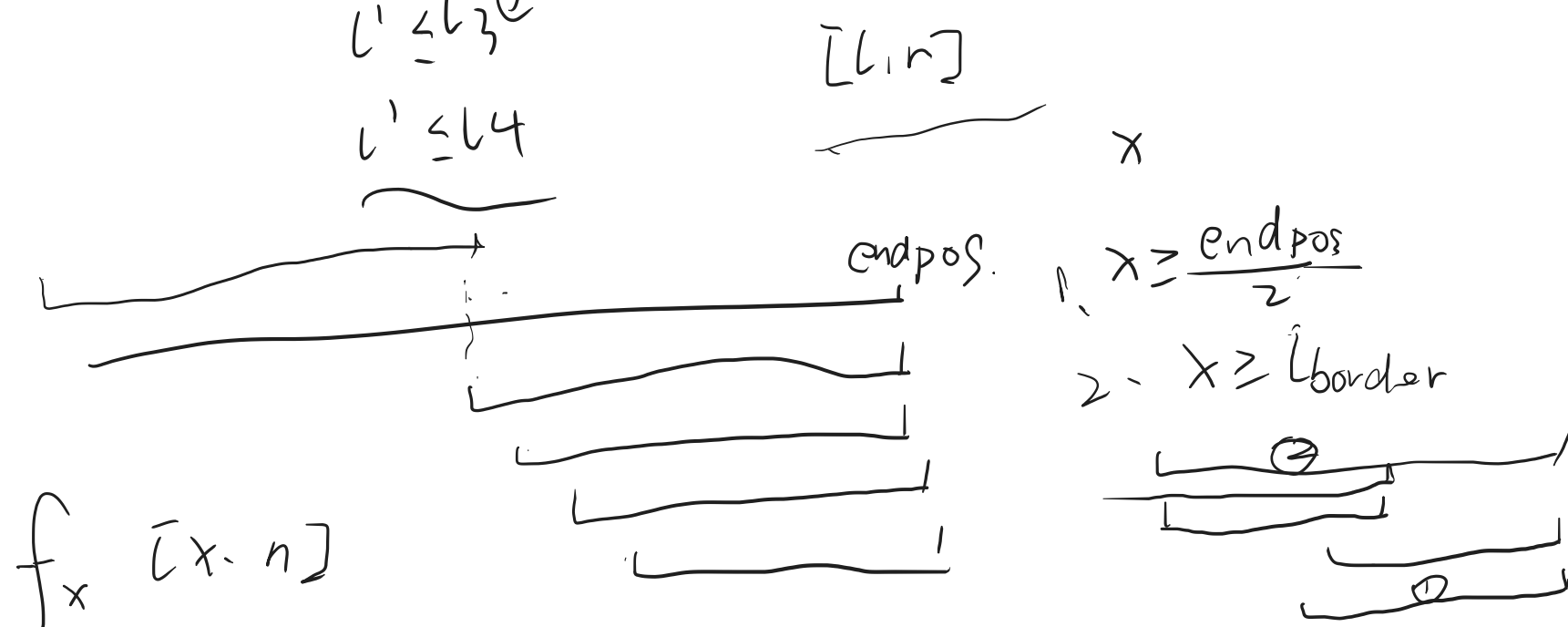
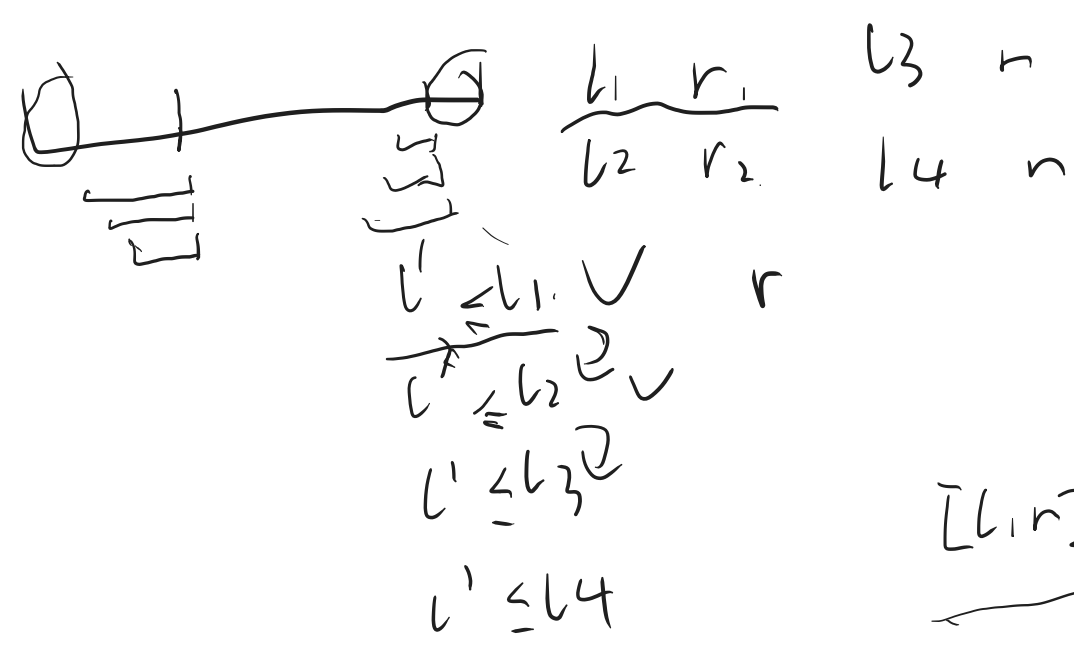
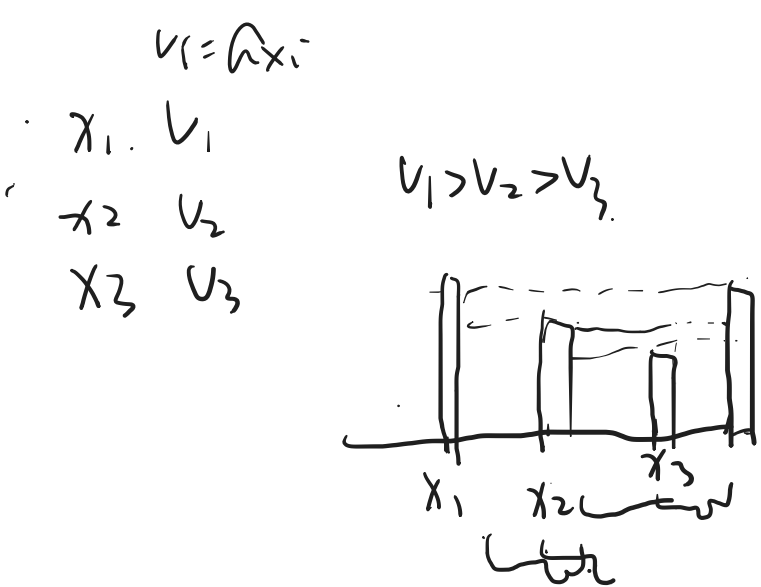
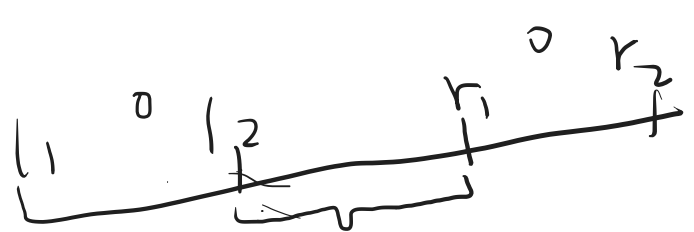


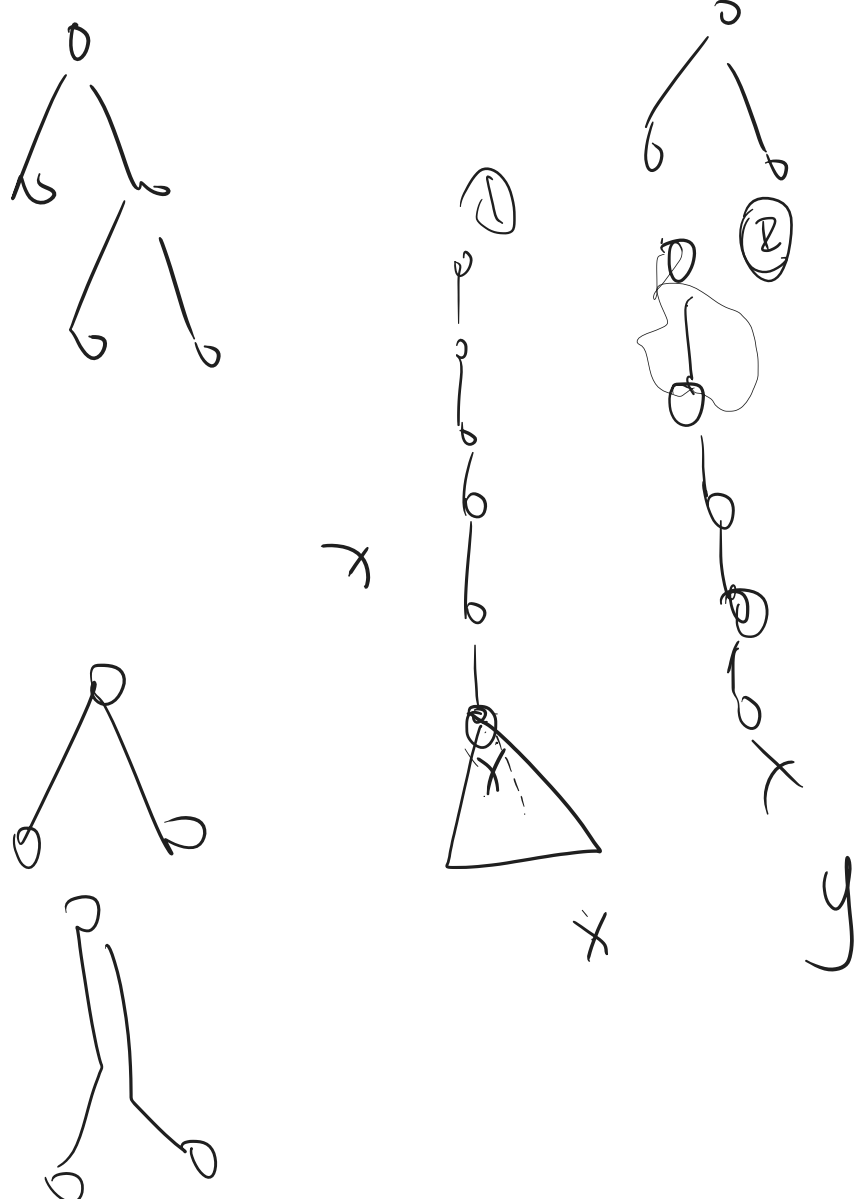
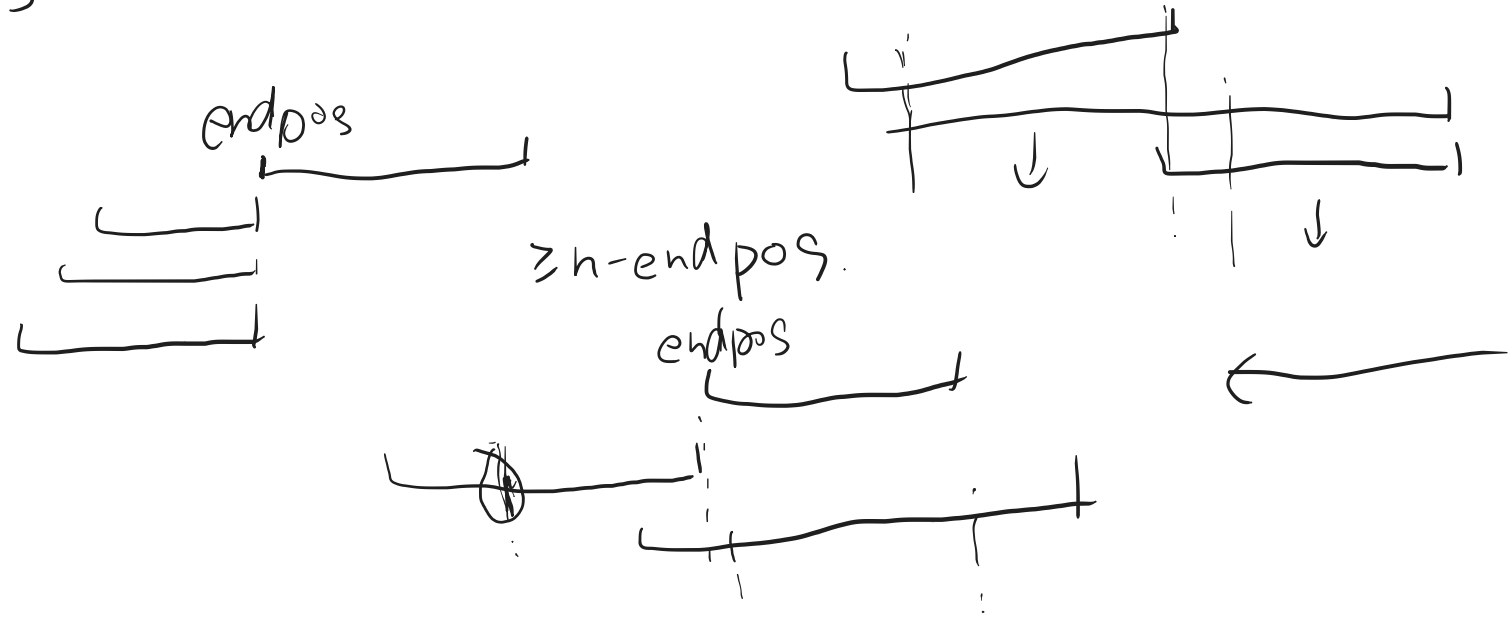


$$l_1 < l_2 < r_1 < r_2$$

$$[l_2, r_1]$$



$$f_x [x, n]$$



step 1: lca  
step 2: x fi  
step 3: lca2  
step 4...